DS-A3: Regression Baseline and Diagnostics

1. Introduction and Dataset

Predict one-year disease progression using all ten original features in the sklearn Diabetes dataset. There are 442 observations and no missing values. Inputs are loaded with scaled=False.

2. Data Splitting and Leakage Audit

An 80/20 split gives 353 training and 89 test rows. Both the split and shuffled five-fold CV use seed 42. StandardScaler is fitted inside each pipeline. Model selection uses training CV MAE before test evaluation.

3. Baseline and Model Training

Compare a mean DummyRegressor, LinearRegression, Ridge, and Lasso. Ridge and Lasso search alpha values 0.1, 1, 10, and 100. All configurations and fold scores are retained in main.ipynb.

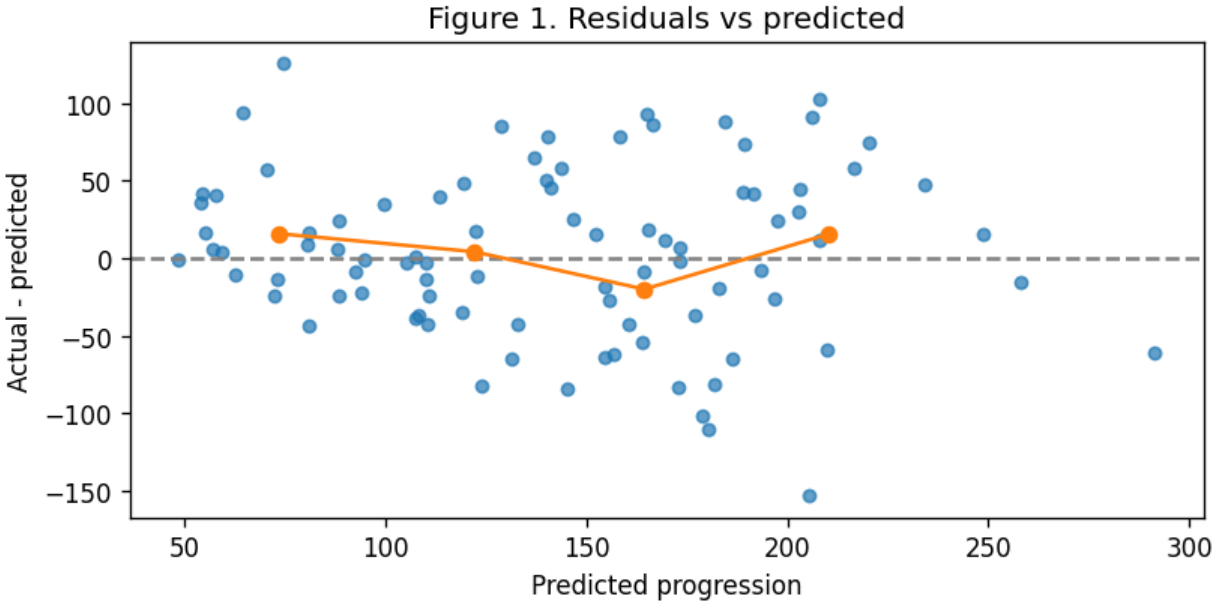
```
data = load_diabetes(scaled=False, as_frame=True)
Pipeline([('scale', StandardScaler()), ('reg', Ridge())])
GridSearchCV(model, grid, cv=cv,
             scoring='neg_mean_absolute_error')
```

4. Model Comparison

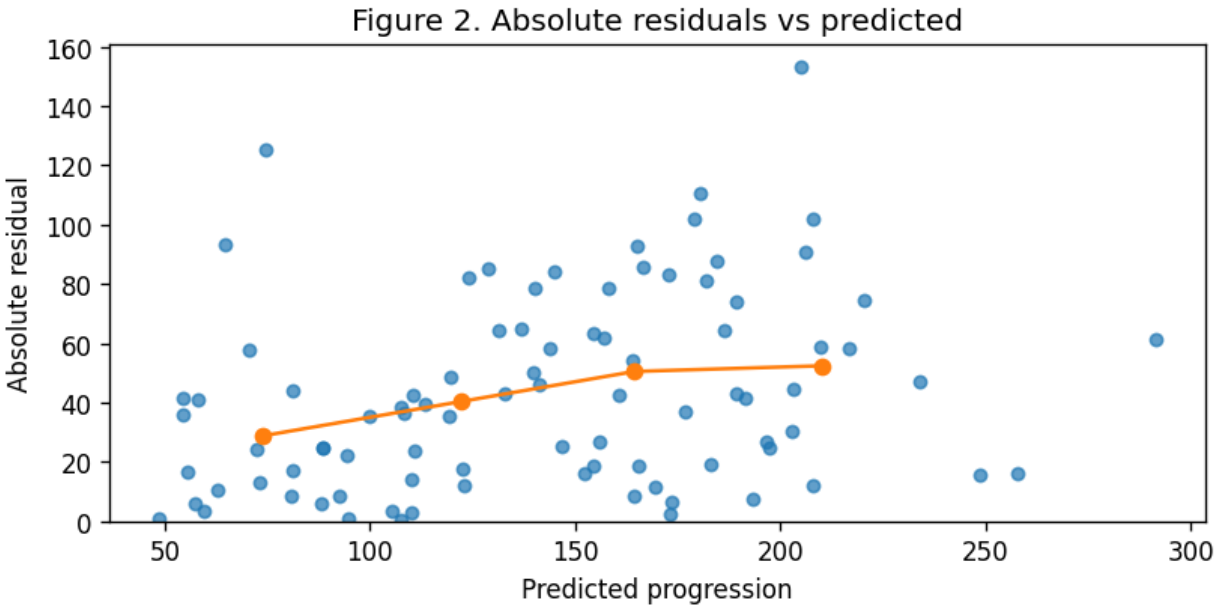
Model	alpha	CV_MAE	CV_SD	Test_MAE	Test_RMSE	Test_R2
Dummy	NaN	66.867	5.632	64.006	73.222	-0.012
Linear	NaN	45.048	1.773	42.794	53.853	0.453
Ridge	1.000	44.999	1.808	42.812	53.777	0.454
Lasso	0.100	44.956	1.837	42.805	53.709	0.456

Lasso (alpha 0.1) has the lowest CV MAE, 44.956. Its gain over Linear is only 0.093, so regularization adds little. Its test MAE is 42.805, RMSE 53.709, and R-squared 0.456. MAE and RMSE are in target units.

5. Diagnostics and Error Analysis



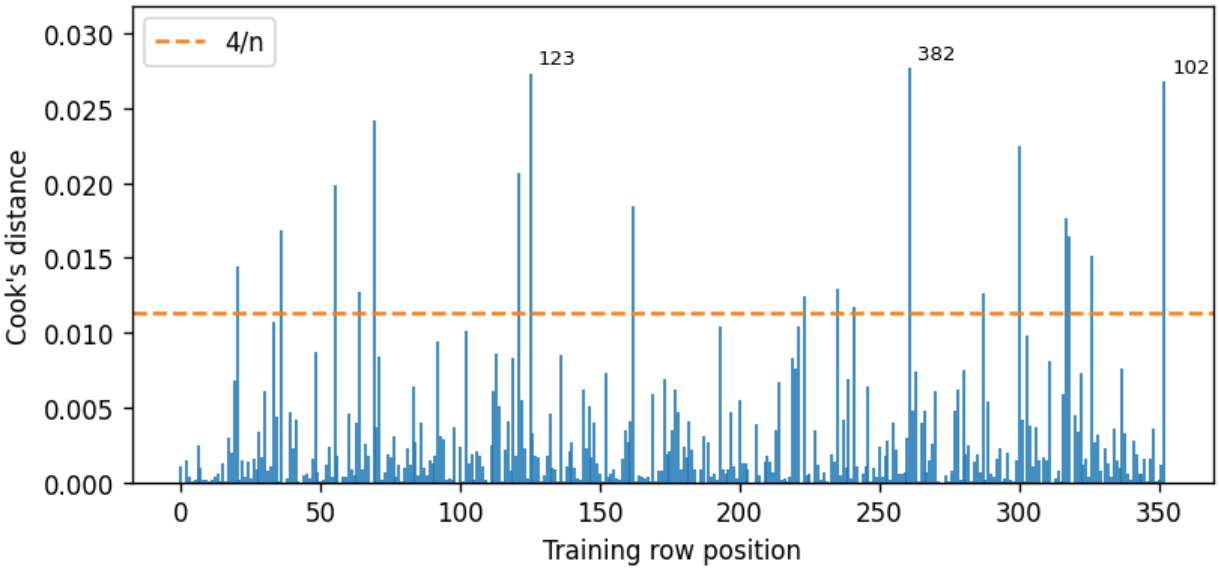
Residual = actual - predicted. The third prediction quartile has mean residual -20.02, showing local overprediction. There is also considerable scatter around zero.



Quartile MAE rises from 28.71 to 52.38 as predictions increase, suggesting possible heteroscedasticity.

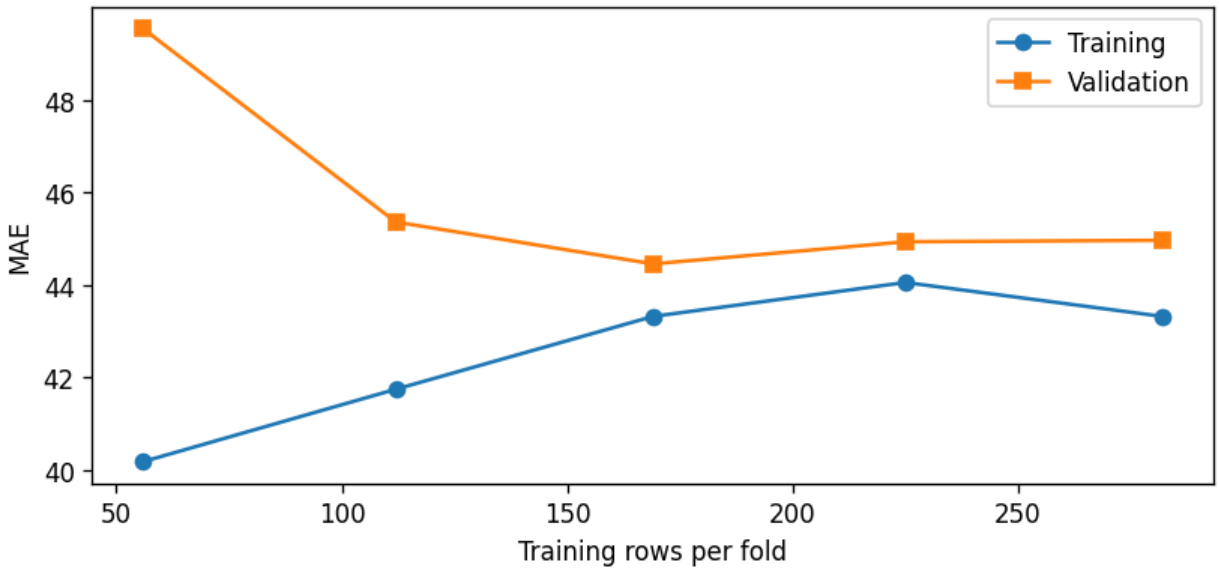
Training Influence and Learning Curve

Figure 3. Cook's distance (training OLS)



Training OLS has 18 rows above the 4/n screening threshold. Row 382 has the largest Cook's distance (0.0277). These rows can shift OLS coefficients; none are deleted.

Figure 4. Learning curve



Using training data and five-fold CV with the selected alpha fixed, validation MAE falls from 49.53 to 44.96 and then levels off. The final train-validation gap is small (1.64), so severe overfitting is not evident.

Critical Slices

BMI groups use the training tertiles, 24.2 and 27.9. Intervals are right-closed.

BMI slice	n	MAE
Low	38	34.490
Middle	23	47.433
High	28	50.288

The high-BMI group is hardest to predict: MAE 50.29 across 28 test rows.

Error Cases

sample_id	actual	predicted	absolute_error	bmi
56	52.000	205.195	153.195	30.200
77	200.000	74.619	125.381	23.000
211	70.000	180.323	110.323	29.800

Rows 56 and 211 are overpredicted despite low actual outcomes; row 77 is underpredicted. The ten inputs do not fully explain individual outcomes.

6. Discussion and Limitations

Lasso wins by CV, but Ridge and Lasso barely improve on Linear. Diagnostics show local bias, possible heteroscedasticity, and larger errors in the high-BMI group. More data may help, but the learning curve is flattening.

Coefficients describe associations, not causal effects: correlated features and unmeasured factors can affect them. This small dataset and single split limit generalization, especially for small subgroups. These predictions are not clinical diagnoses.

7. Reproducibility

Use Python 3.12 and requirements.txt, then restart the kernel and run main.ipynb from top to bottom. The notebook recreates results.csv. Python 3.12.14 and sklearn 1.7.2 were used for this run. Command-line instructions are in README.md.

Data source: scikit-learn.org/stable/datasets/toy_dataset.html#diabetes-dataset